

Word count: 2109

An investigation on Aspirin hydrolysis

Research Question: How temperature affect the rate of aspirin breakdown in an acidic solution in relation to pH change over time?

Introduction

In chemistry, hydrolysis can be defined as the process whereby water reacts to break the bonds of compounds, and bases and acids help as catalysts. In the case where hydrolysis occurs

because of acid catalysis, the hydrogen ions help break the bonds of the compounds, hence the formation of molecules of water (Alibrandi et al., 1996).

Background information

The ester, acetylsalicylic acid, Aspirin, reacts with water to produce acetic and salicylic acids, and it is an analgesic and anti-inflammatory agent (WebQC.org, 2025). The hydrolysis reaction for Aspirin can be written as follows:



The process also accelerates with higher temperatures: heat increases the energy of molecules; thus, the number of effective collisions between reactants increases too. During the reaction, the pH of the solution gradually decreases due to the acidic products of hydrolysis being produced. It is, therefore, possible to measure the rate of hydrolysis by following this pH change without complex apparatus in school.

It is important to study the dependency of the rate of hydrolysis on temperature, as it simulates how drugs can behave in human bodies under different conditions. The rates at which aspirin breaks down can affect how quickly the drug activates or how long it stays active under warmer conditions (Delattre et al., 2025). In addition, looking into the temperature dependence of aspirin hydrolysis helps show the principle that molecules gain kinetic energy with heat, which

is also important to areas where temperature control is key. (e.g., pharmaceuticals, and more broadly within any chemical processes).

This experiment examines how rising temperatures affect the aspirin hydrolysis rate by measuring changes in pH as time progresses. Based on background research and theory, I hypothesize that with increased temperature, there is an increased rate of aspirin hydrolysis, hence faster and larger drops in pH. At a higher temperature, there is a higher collision frequency, hence increasing the acid-catalyzed hydrolysis of aspirin.

Methodology/ Research design

Independent variables:

Temperature of reaction: 30°C, 40°C, 50°C, 60°C, 70°C. The temperature was controlled within a thermostatic water bath (± 0.5 °C); the solutions were heated for 2 min before the measurement.

Dependent variables:

pH as a function of time and aspirin hydrolysis rate by means of a pH probe previously calibrated (± 0.01 pH).

Controlled variables

Volume of aspirin solution dispensed by a 10mL volumetric pipette: 10.00mL. Volume affects the value of the concentration and the hydrolysis rate.

Volume of 1.0M HCl (10.00mL) delivered using a 10mL pipette. Controlled because the concentration of acid affects the pH and the rate of catalysis.

The pH probe was calibrated with pH 4.00 and pH 7.00 buffers before every temperature set.

This is to ensure that the pH readings to be taken are accurate for the calculation of the rate.

All the runs were carried out with the same type of reaction container, of 20mL test tubes since different surface areas affect the rate of reaction through heating.

Method

For this experiment, I decided to investigate how temperature affects the rate of aspirin hydrolysis while also keeping other factors constant. So, for each trial, the solutions would have a varying temperature while the aspirin concentration, HCl concentration, reaction time, and total volume were kept constant. By isolating temperature to be the independent variable, I can treat any differences in reaction rate directly linked to temperature effects.

The aspirin solution was made by crushing 5 g of aspirin tablets with a mortar and pestle, then to dissolve the measured mass of aspirin, 100 mL of distilled water was added, mixed, and filtered the final mixture.

The hydrochloric acid solution was also prepared in the same way by measuring 100 mL of 1.0M HCl into a beaker and tested with a calibrated pH probe to reduce inaccuracies. For each trial in the experiment, 10ml of the aspirin solution is mixed with 10ml of HCl in a test tube, then is

placed and set in a water bath set to the desired temperature. The pH is continuously recorded by using a pH probe in the test tube and sent to the data logger device, such as a phone, allowing the change in acidity to be recorded over time to measure the rate of reaction.

For the experiment, each temperature condition was repeated five times with fresh solutions to reduce anomalies. All other conditions, for example, probe calibration, stirring rate, and reaction volume, were kept constant. This allowed me to use an initial slope of the pH vs time curve to be used as a measure of the rate of hydrolysis.

Safety Precautions

During the experiment, protective equipment such as goggles and lab coats was always worn.

Hazards	Risk	Precautions
High temperature water bath	Burn risk from hot water and heated glassware	handling test tubes with tongs
Hydrochloric acid	Risk of skin and eye irritation	safety goggles and lab coats were worn

electrical equipment by water	Potential hazard is contacted with water	equipment were carefully positioned to keep dry
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Data collection

Raw data

The pH was recorded from 0 seconds to 600 seconds and the change in pH was noted down in an Excel sheet.

Temperature (°C)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Mean pH
30	0.14	0.13	0.15	0.14	0.13	0.14
40	0.07	0.18	0.22	0.20	0.19	0.17
50	0.24	0.28	0.27	0.25	0.29	0.27
60	0.31	0.33	0.48	0.45	0.39	0.39
70	0.65	0.61	0.61	0.60	0.57	0.61

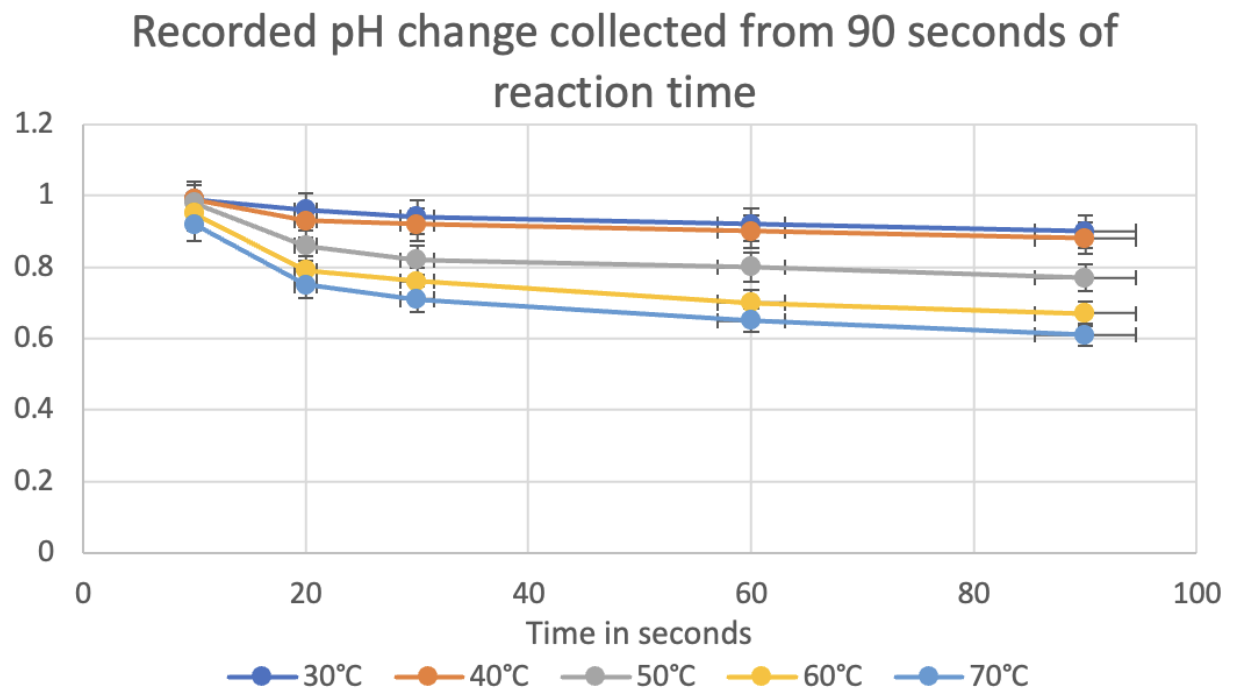
Processed data

First, the average pH change over the first 90 seconds at each temperature is calculated. The mean pH at each time was calculated by averaging the five trials for each temperature. These averages can be used to lower random errors and provide better and more reliable data to analyze. As shown in the chart below, the pH decreases due to the formation of salicylic and acetic acid, showing the hydrolysis of aspirin.

Temperature	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average	Uncertainty	%uncertainty
30	0.14	0.13	0.15	0.14	0.13	0.14	0.01	10.7
40	0.07	0.18	0.22	0.20	0.19	0.17	0.08	10.7
50	0.24	0.28	0.27	0.25	0.29	0.27	0.03	10.7
60	0.31	0.33	0.48	0.45	0.39	0.39	0.09	10.7
70	0.65	0.61	0.61	0.60	0.57	0.61	0.04	10.7

Uncertainty was calculated by combining the major uncertainties — pH probe ± 0.01 (~1%), temperature stability ± 0.5 °C (~2%), volumetric error (~3%), and probe errors, mixing effects (~8%) — gives a total uncertainty of approximately $\pm 10\%$.

pH vs time graph



Here is a visual showing the rate of the pH drop at different temperatures

X-axis: time in seconds (10-90)

Y-axis: mean pH

The graph above shows the downward trend through a time-pH diagram. The highest drop in pH happens in the first few minutes and gradually becomes a lower curve. This shows how the hydrolysis reacts the quickest at the beginning and slows down as aspirin reacts. For example, the pH drops most rapidly from 1 to 0.8 in the first 20 seconds overall, whereas the difference in pH between 90 to 120 becomes much more gradual. This difference in gradients shows decreasing reaction rate over time.

Processed Data Discussion

In conclusion, the graph of the curve shows that the slope of the curve is more evident when the temperature is raised, showing that the pH change is related to temperature. The blue line (50°C) shows that it has a bigger pH change than the purple line (40°C), and further observation of the dark green line shows that it has the steepest descent in pH change, showing that temperature and reaction rate are related. All this points to the observation of the collision theory, which states that higher temperatures correspond to higher movement and collision of molecules ("How Temperature Affects the Rate of a Reaction | the Chemistry Blog").

Generally, the results of the processed data indicate that temperature is one of the factors that greatly affect the rate of aspirin hydrolysis reaction, whereby the higher the temperature, the faster the reaction.

From the above analysis, the relationship between temperature and the rate of aspirin hydrolysis is well described. Upon increasing temperatures, the decline in pH values occurred sooner due to faster reaction rates. The steepest point of the curve represented in the graph lies at 60 °C, compared to 30 °C, which had the lowest rates. This depicts the same concept that at higher temperatures, kinetic energy increases, causing faster decompositions of aspirin.

In conclusion, the findings are sufficient evidence that the rate of the reaction of aspirin hydrolysis is temperature dependent. Despite some limitations that may influence the accuracy of the findings, the trend is quite evident and supported by chemistry laws and theories. Even though there were some uncertainties, the errors aren't significant enough to affect the validity of the data since the trend in the results is still clear.

Evaluation

Even though there are clear correlations between my experiment data and the research hypothesis, there were several limitations that impacted the accuracy of the recorded rate values. One main source of uncertainty was the pH probe. Even after calibration, the probe still showed minor drifts in recordings after long durations, and readings would fluctuate when the stir bar was moved near the sensor. Since my reaction rates were directly calculated from my pH changes, even small changes in the recordings can affect my results. Measuring pH in a closed environment with a stable pH holder would increase my precision.

The temperature control also introduced errors. The water baths maintained set temperatures, but there were small shifts in temperature ($\pm 0.5^\circ\text{C}$) whenever the lid of my water bath would be open. Because my experiment is very temperature sensitive, even a small change in degrees can change the reaction rate. To improve accuracy, the water bath should be in a closed environment with less movement on the lid or equipment.

Another limitation is from the evaporation during heating. As the mixture was heated in open test tubes within the water bath, small amounts of solvent may have evaporated, especially at higher temperatures. This will increase the ratio concentration of aspirin and hydrochloric acid, which can cause the hydrolysis rate to appear higher at increased temperatures. Since evaporation increases with temperature, this creates a systematic error that could exaggerate the observed relationship. Using sealed reaction tubes or covering the tubes with parafilm can improve accuracy and reduce solvent loss.

Another limitation is from the loss from transfer of liquids from different beakers, such as pouring solutions into the test tubes; a small amount of solution will remain on the beaker due to adhesion. This resulted in variations in actual volumes in each trial. These losses happen inconsistently between trials, therefore introducing random error, reducing the reliability of the calculation rates. Conducting reactions directly in volumetric containers would reduce this source of error.

Conclusion

There is substantial evidence that temperature does in fact exhibit a specific effect on the rate of hydrolysis in an acidic solution. Even in spite of the fact that the experimental limitations have raised doubts, it is certain that the trend has remained unchanged in every trial: the higher the temperature, the higher the initial pH level and the rate of hydrolysis. In any case, the results are remarkably similar to that which had been expected concerning the effect of ester hydrolysis in an acidic solution. Despite the limitations that have been identified, my observations do verify that temperature makes an impact on the aspirin hydrolysis process.

However, the reliability of this conclusion is strengthened by the definiteness and recurrence of this pattern for every temperature, considering both systematic error such as miscalibration of the probe, and random errors, such as minor differences in mixing between trials. Because the pH-time curves for each replicate are similar, the correlation between temperature and reaction rate is also generally valid. However, the actual values of rates measured are less

reliable due to problems like drift in temperature probes, imprecision in volume measurements, and incomplete equilibration. In conclusion, this experiment confirms that temperature does affect the rate of aspirin breakdown in acidic solutions.

Further Investigations

Further research could be comparing how the rate of aspirin breakdown could differ when in alkaline solutions, instead of previous acidic solutions, to determine how different solutions affect aspirin breakdown.

References

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